

Advanced Simulations: Surveys

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Outline

We have prepared four sets of simulation scripts that can be run and modified for this hands-on session. The sets can be found in the **surveys** directory in the GitHub repository:

- **chandra_deep_field**

Create a simulation of the Chandra Deep Field South (CDFS) with the NewAthena WFI.

- **dithering**

Create a simulation of the satellite in dithering mode and generate an exposure map of the observation.

- **simple_slew**

Run an all-sky survey simulation for NewAthena WFI for multiple sources.

- **arf_generation**

Run an ARF correction using `sixte_arfgen`.

Where to find the files

Clone the workshop GitHub repository:

https://github.com/stierhofj/Sixte_Workshop_2026.git



simulation_examples / surveys → ..

📁 arf_generation
📁 athena_wfi_cdfs_deep_field
📁 dithering
📁 simple_slew
<i>Exercise sets</i>

📁 input	<i>Extra simulation input files</i>
📄 01_create_simput	<i>Scripts of the chosen exercise folder</i>
📄 02_slew_attitude	
📄 03_simulate	
📄 04_imgev	
📄 05_chip_image	
📖 README.md	<i>Exercise information</i>
🐍 chip_image.py	<i>Additional Python scripts</i>
📄 run	<i>Runs all the scripts at once</i>
📄 setup.sh	<i>Setup exercise parameters</i>

Start running the exercises

Feel free to start from any folder; however, do not forget to read the README.md.

Let us know if you have any questions!

athena_wfi_cdfs_deep_field: simulate the Chandra Deep Field South

dithering: simulation of dithering mode and exposure map generation

simple_slew: All-sky survey simulation

arf_generation: ARF correction